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Guidelines for Technical Presentations

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1. INTRODUCTION

The presentation of technical work is an integral part of any engineer's job. Students therefore have to learn to deliver technical presentations and this document aims to provide them with guidelines for the preparation and delivery of technical presentations. Since several comprehensive guides on this subject are available, only the main points are summarised here.

The circumstances of two particular groups of speakers are also specifically addressed: final year project students (i.e. students who deliver presentations as part of their Mechanical or Mechatronic Project) and postgraduate students (i.e. students who deliver public presentations as part of their oral examinations).

2. PREPARATION

Thorough preparation is one of the most effective ways of overcoming the fear of presentation.

2.1. The aim

The speaker must have a clear view of the purpose of the presentation. The shorter the presentation, the more important it becomes to keep to matters directly related to the purpose.

In the case of final year project students, the first presentation is about the progress to date. It is primarily aimed demonstrating to the project's clients that the speaker has a clear understanding of the project's objectives, that good progress has been made to date and that the speaker is competent to complete the project by the due date. The second presentation for final year project students is a close-out report, in which the results achieved in the project are presented. The presentation must also demonstrate that the results were achieved in a professional manner and can therefore be trusted.

In the case of postgraduate students, the aim of the public presentation is to describe the relevance of the research that was performed (in other words the research objectives and motivation) and to show that the most important conclusions follow logically from the results of thorough scientific investigation.

2.2. The audience

Determine:

- Who is the audience?
- What is their expectation of, or interest in, the presentation?
- What is the level of their technical expertise?

Do not attempt to explain the Navier-Stokes equations to an accountant in 5 minutes, but do not give a technical manager a superficial presentation either.

The target audience for a final year project presentation is the project's clients and their technical level is usually that of a graduate mechanical or mechatronic engineer. The target audience for the postgraduate students' presentations is the lecturers of the Department and other postgraduate students. This audience has a broad knowledge of the different disciplines within mechanical and mechatronic engineering, but not on the same level as the student's supervisors and examiners.

2.3. Duration of presentation

The shorter the presentation, the better the preparation must be. Be very selective about what you would like to convey. Do not attempt to cover everything if you do not have sufficient time. Choose only the most important aspects to talk about.

As a broad guideline, allow about 1 to 3 minute per slide. It is better to finish slightly early, rather than rush to finish or stop before the presentation is complete.

2.4. Language

Use either English or Afrikaans. Do not use a combination of languages. From time to time a speaker may use a technical term in another language during the presentation, but be careful not to use primarily English terminology in an Afrikaans presentation.

Be careful when choosing your words. The precise and accurate choice of words increases the presentation's credibility and competence in the eyes of the audience, particularly regarding technical terms.

It is of the utmost importance that no spelling or grammatical errors appear in the presentation. If the client/audience/examiner sees that the speaker cannot even get something as simple as spelling correct, how can he/she be trusted with important technical work? Proof read the presentation on paper, since it is well known that one will notice errors on paper far more easily than on a computer screen.

2.5. Slides

PowerPoint, or software, slides are common presentation aids. The minimum expectation nowadays, is that a presentation will use slides.

Use the following as the general rule for the amount of text on a slide:

- Do not explain more than one idea per slide.
- Do not use more than seven words per line.
- Do not use more than seven lines per slide.

Guidelines for layout:

- Avoid using backgrounds that will distract the audience's attention, such as overpowering graphics or animations. Further, animations of the slide's elements should be used very selectively, to not distract or irritate the audience.
- Ensure that there is sufficient contrast between the text and the background. The text should still be legible on the projected screen even if the room is not particularly dark. Remember that the colour contrast of a data projector is weaker than that of a computer screen. Colours like cyan and yellow can often not be distinguished against a white background.
- The minimum text (font) size on a slide is 18pt. Aim for 24pt or larger.
- Simplified schematic representations or figures are often much better than text on a slide. (Confucius: *one picture is worth a thousand words*). Digital cameras make it easy to use photographs, but aim to show only relevant information in the photograph, for example by choosing a clean background. It is important to add labels to figures and photographs, so that everything that will be referred to in the presentation is indicated.
- Video inserts can be very effective, but are also very vulnerable as it may not play on some laptops or data projectors.
- Graphs should preferably not have more than three lines. The labels and text must be clearly legible. Lines must be thick enough to be clearly displayed on data projectors with low resolutions.
- Avoid tables with more than two rows and two columns.
- Place parameters that need to be compared, on the same slide and in the same graph/table if at all possible.
- Be careful not to use too many different colours on one slide. Be especially careful for colours which give low contrast, e.g. yellow.
- Do not use more than one font type on a slide. Be careful with fonts such as Times New Roman, as the thinner parts of the letter are not always visible on certain data projectors. Arial displays well on most data projectors.

2.7. Additional aids

Remember to have a pointer at hand, to point to specific elements on slides. Do not use the shadow of your finger, but rather use the mouse instead.

The use of models and physical examples can be very effective in capturing the audience's attention and to facilitate the understanding of key concepts. Beware of models that are too small or too complex. Also take into consideration the time it would take to get the bigger models into position during the presentation. Too many aids can confuse the audience and can take too long to discuss with the audience.

Some speakers find it easier to use presentation notes (cue cards) during the presentation, while others prefer to speak from memory. Should you choose to speak without notes, ensure that there is a cue on the slide for each aspect that should to be mentioned.

2.8. Structure of the presentation

Ensure a logical flow during the presentation. The audience must always be “with” the speaker and where the presentation is going. The presentation must always have an introduction, a body and a conclusion.

2.8.1. Introduction

- Introduce the speaker and topic.
- Give an overview of the presentation.
- Give the background to the topic. The remainder of the presentation will not make sense to the audience if they do not understand the context.
- Provide a clear problem statement and description of the objectives of the work.
- Give a summary of the approach or strategy that was or will be used to reach the goals.

2.8.2. Body

- Focus in particular on discussing the results. Just enough information about the methods applied must be given to demonstrate to the audience that the conclusions reached in the summary are plausible.
- When presenting project/research proposals, group activities into three or four main phases. These phases must be briefly explained and accompanied by a Gantt chart, showing the order and duration of the activities. A budget summary should also be presented. Be careful not to go into too much detail about the phases or budget. Only the main points should be mentioned.
- The order of the presentation's content normally does not follow the order in which the work was done, but rather builds an argument that leads logically to the conclusions.

- Theoretical assumptions and developments can rarely ever be fitted into a presentation. Rather just give the point of departure, assumptions, approximations and result.

2.8.3. Conclusion

A decided conclusion is essential. Do not just say that a background and motivation was given, but rather give the key aspects that were discussed in each section. It must bring together all the important points that you would like the audience to remember.

State especially the conclusions that were drawn, highlight the importance of the main findings and answer the question "*so what*". If you have recommendations for further work, make them here, but remember that the last part of the presentation is usually remembered best. Therefore, try not to end the presentation with recommendations for improvement. Rather close with a discussion of the value of the results, i.e. the "*so what*".

Do not close with "Thank you. Any questions?" At most presentations, a chairperson will preside and offer the audience the opportunity to pose questions. Then it would be inappropriate for a speaker to invite questions.

2.9. Dry-run

NB: Test the computer and data projector thoroughly before the presentation. Leave enough time to fix any problems. Ensure that the computer you are going to use during the presentation has compatible software and version, particularly if the software you used to prepare the presentation was released in the last three years.

It is important to become familiar with the environment in which the presentation will take place, e.g. the data projector operation, what the slides look like from the audience's perspective, how to operate the lights and speakers, etc. A practice presentation in an environment that is as close to the actual situation as possible would be good preparation for speakers with little or no experience.

For inexperienced speakers, it is a good idea to make a video recording of the practice presentation and to evaluate it. It is usually very informative to see such a recording, because one often does or says things that you are not aware of.

3. PRESENTATION

The speaker must, before he/she starts to speak, make sure that all the presentation aids are ready. It is better for the audience to wait a few seconds before the presentation begins, than to have the presentation interrupted after a few seconds, for example, while the speaker searches for a pointer. The slide that introduces the speaker and subject can be displayed while the aids are prepared. If presentation notes or cue cards are used, the speaker must make sure that there is a convenient place to put them down as soon as he/she has finished with each page or card.

The speaker must be careful not to stand in front of the projector's beam.

A speaker should never read from a slide (there should in any case not be many complete sentences). Reading from your notes or cue cards also does not create a good impression, but it may be the best way out for speakers that feel very uncertain or cannot speak comfortably without them.

Maintain as much eye contact with the audience as possible. If a computer presentation is used, look at the computer screen and not the projector's screen whenever necessary.

Use formal language, but a joke or two in good taste can also be appropriate. The language used must reinforce the impression of professionalism.

The speaker should come across as being confident, sincere and enthusiastic. Language, tone of voice, facial expressions, gestures and body language all contribute to the impression you create. Be careful not to use a monotonous tone. Appropriate use of volume and tempo, with suitable hand gestures, is very important. Speak clearly and slowly (but not too slowly). Try not to "um" and "ah". This indicates that you are still busy speaking and want to add more. Try to remember that your presentation is your turn to speak, and that you will not be interrupted.

Point on the slide where you would like the audience to focus their attention. Do not indicate so quickly that the audience will miss it if they blink, but try not to show them how shaky your hand is. If you are using a computer presentation, use the mouse as pointer or a laser pointer.

When a graph (chart) is used:

- Broadly explain what the graph is about
- Explain each axis
- Explain each line
- State the conclusions that should be drawn from the graph

When a figure, table or graph is displayed, it is essential that the speaker leads the audience in its interpretation. Even if the conclusions seem self-explanatory to the speaker, the audience is seeing the figure, table or graph for the first time and can therefore not make the deductions immediately.

Be careful of not speaking in the second person (e.g. not "To determine the pressure recovery, you should use your pressure transducer..." , but rather " To determine the pressure recovery, the pressure transducer is used...").

4. CONCLUSION

This document provides a number of guidelines that need to be kept in mind during the preparation and delivery of technical presentations. If a speaker is well prepared and certain about his/her case, then he/she can relax and deliver the presentation with confidence. The worst that could happen is that someone in the audience disagrees with the speaker, and even that is normal.